INCH - POUND

MIL-A-47318A (MI)
15 October 1990
SUPERSEDING
MIL-A-47318 (MI)
10 July 1987

MILITARY SPECIFICATION

ADHESIVE, COPOLYMER POLYURETHANE

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of a two-part epoxy polyurethane copolymer adhesive.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8040

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 790	-	Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials, Standard Test Methods for
ASTM D 1002	-	Strength Properties of Adhesives in Shear by Tension Loading (Metal-to-Metal), Standard Test Method for
ASTM D 1084	-	Viscosity of Adhesives, Standard Test Methods for
ASTM D 2240	-	Rubber Property - Durometer Hardness, Standard Test Method for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article inspection. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.2 and 6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

3.2 Material. Material shall be an epoxy copolymer urethane adhesive base (part A) and a hardening agent (part B), in separate containers clearly indicating material (see 4.6.1).

3.3 Cure time. The material shall cure in not more than 4 hours at a temperature of 65 ± 5° Celsius (C) (see 4.6.2).

3.4 Pot life. Pot life at 25 ± 2°C shall be not greater than 1 hour (see 4.6.3).

3.5 Viscosity. Immediately after mixing, the viscosity shall be 2000 to 4000 centipoises (see 4.6.4).

3.6 Hardness. Shore D hardness shall be not less than 70 at 25 ± 5°C (see 4.6.5).

3.7 Shear strength. Lap shear strength shall be not less than 2600 pounds force per square inch (lbF/in^2) (see 4.6.6).

3.8 Flexural strength. Flexural strength shall be not less than 8500 lbF/in^2 (see 4.6.7).

3.9 Container size. Unless otherwise specified in the contract or order (see 6.2), the container sizes and designators shall be: ounce (OZ), pint (PT), quart (QT), and gallon (GL).

3.10 Shelf life. When stored in unopened containers at a temperature not greater than 25°C for a period of 1 year, the material shall meet all requirements of this specification (see 4.6.8).

3.11 Marking for identification. Marking for identification shall be in accordance with MIL-STD-130 and shall include, but not be limited to, the following (see 4.6.1):

a. Title, number, and date of this specification

b. Military part number

c. Manufacturer's name or trademark

d. Any necessary hazard warnings

e. Mixing instructions

f. Component identification (part A or part B)

g. Date of manufacture.

3.12 Workmanship. The material shall be free from lumps, foreign matter, or other visible defects that would detract from its appearance or function (see 4.6.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed under ambient conditions.

4.4 First article inspection.

4.4.1 First article sample. When specified in the contract or purchase order (see 6.2), a first article sample shall consist of one lot of adhesive manufactured in one production batch run from one process and sufficient to perform all tests specified herein. The first article sample shall be fabricated using the same processes, facilities, and materials as are proposed for production. The first article shall meet all requirements of this specification.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of all the tests listed in table I and as specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

TABLE I. Quality conformance inspection.

Examination and test	Requirement paragraph	Method paragraph
Material	3.2	4.6.1
Cure time	3.3	4.6.2
Pot life	3.4	4.6.3
Viscosity	3.5	4.6.4
Hardness	3.6	4.6.5
Shear strength	3.7	4.6.6
Flexural strength	3.8	4.6.7
Container size	3.9	4.6.1
Shelf life	3.10	4.6.1
Marking	3.11	4.6.1
Workmanship	3.12	4.6.1
Packaging	5	4.6.1

4.5.1 Lot size. Lot size shall consist of all the adhesive manufactured in one production batch run from one process forming part of one contract or order and submitted for acceptance at one time.

4.5.2 Sampling. Unless otherwise specified, sampling shall be performed on a quantity of adhesive sufficient to assure that all requirements of this specification have been met.

4.6 Test methods.

4.6.1 Visual inspection. Visual inspection shall verify compliance with 3.2, 3.9, 3.10, 3.11, 3.12 and 5.

4.6.2 Cure time. Mix an aggregate of 1 ounce of adhesive and expose to a temperature of 65 ± 5°C for a period of 4 hours. Successful completion of the hardness test (specified in 4.6.5) shall assure compliance with 3.3.

4.6.3 Pot life. Mix a sample as specified in 4.6.2. Maintain at 25 ± 2°C for a period of 1 hour. If hardness of the material precludes spreadability, requirements of 3.4 have been met.

4.6.4 Viscosity. Viscosity shall be determined in accordance with ASTM D 1084 to assure compliance with 3.5.

4.6.5 Hardness. Shore D hardness shall be determined in accordance with ASTM D 2240 to assure compliance with 3.6.

4.6.6 Shear strength. Shear strength shall be determined in accordance with ASTM D 1002 to assure compliance with 3.7.

4.6.7 Flexural strength. Flexural strength shall be determined in accordance with ASTM D 790 to assure compliance with 3.8.

4.6.8 Shelf life certification. The supplier shall submit certification of shelf life capability to assure compliance with 3.10.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

4.7 Inspection of packaging. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with the requirements of MIL-STD-1190 and MIL-STD-129.

5. PACKAGING

5.1 Packaging and packing. Packaging and packing shall be level C in accordance with MIL-STD-1190 (see 4.6.1).

5.2 Marking for shipment and storage. Marking for shipment and storage shall be in accordance with MIL-STD-129. In addition to the requirements of 3.10, marking shall include, but not be limited to, the following:

a. Lot or batch number

b. Manufacturer's grade, class, or part number as applicable

c. Storage requirements.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The material covered by this specification is intended for use in various rocket motor applications.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.1.1)

c. The military part number

d. Whether a first article sample is required (see 3.1, 4.4, and 6.3)

e. Inspection conditions if different from 4.3

f. Sampling plan if at variance with 4.5.2

g. Quantity desired, by container size

h. Place of delivery.

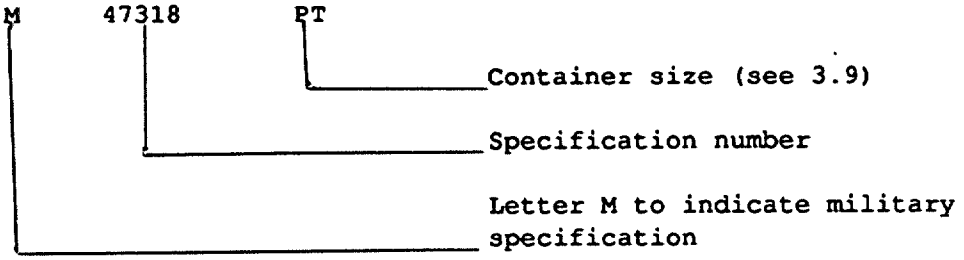
6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Military part number. The military part number shall consist of the letter M, plus the basic number of this specification, followed by a combination of letters denoting container size (see 3.9).

Example: M 47318 PT



6.5 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.6 Subject term (keyword) listing.

Epoxy
Glue
Polymer
Urethane

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Preparing activity:
Army - MI

(Project No. 8040-A160)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-47318A (MI)	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE
ADHESIVE, COPOLYMER POLYURETHANE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER U.S. ARMY MISSILE COMMAND	b. TELEPHONE (Include Area Code) (1) Commercial: (205)876-6980 (2) AUTOVON: 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD FORM 1426

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.